Contribution Journal

Introduction

For this project, my contribution focused on working through and understanding the Exploratory Data Analysis (EDA) of the Titanic dataset. My role was to examine the data, run the provided analysis, interpret the results, and identify the main patterns shown by the statistics and visualizations. I also focused on understanding why each step of EDA is important before developing a machine learning model.

My Specific Contributions

I worked with the Titanic dataset by loading, inspecting, and analyzing the data. I used descriptive statistics to understand numerical information and examined survival patterns based on passenger class, gender, and age. I found that more passengers died than survived, while females and passengers in higher classes generally had better survival rates. I also interpreted the graphs and statistics and explained the main findings in my own words.

My Learning and Reflection

Through my contribution, I developed a better understanding of how EDA is used to investigate a dataset before modeling. I learned that numerical summaries are useful for understanding the general characteristics of data, but visualizations make relationships and patterns much easier to recognize. For example, the passenger-class and gender visualizations made the differences in survival immediately clear. I also learned that missing values must be identified early because data quality can affect later analysis.

Another important part of my contribution was interpreting the results rather than simply producing graphs. The analysis showed associations between survival and variables such as passenger class, gender, and age. I understand that these relationships should be interpreted carefully and should not automatically be treated as proof that one variable caused another.

Limitations and Future Contribution

In future project work, I would strengthen my contribution by creating additional visualizations independently, testing relationships that are not already provided, and documenting the findings clearly for the team.

Conclusion

Overall, my contribution to the project was centered on exploring the Titanic dataset, understanding the EDA process, interpreting descriptive statistics and visualizations, and identifying meaningful patterns in passenger survival. This work helped me improve my Python-based data-analysis skills and my ability to communicate what data findings mean.